Decode a resistor in three rows

NEW EXAMPLE • Green - Blue - Red - Gold • 4-band resistor

Required for every problem, in every grade: show all three rows below. Use your assignment’s own colors and values.

Read left to right; gold is last.

1 Set up the three rows

Green	Blue	Red	Gold
1st digit	2nd digit	Multiplier	Tolerance
5	**6**	**×100**	**±5%**

ROW 1: COLORS

Write the band colors in their order.

ROW 2: JOBS

Identify what each band represents.

ROW 3: VALUES

Match each color and job to the chart.

2 Combine, multiply, bring down

Combine the digits: 5 and 6 → 56

Multiply: 56 × 100 = 5,600 Ω

Bring down tolerance: 5,600 Ω ±5%

JOIN THE DIGITS

Join the digits.

USE THE MULTIPLIER

Multiply 56 by 100.

KEEP THE TOLERANCE

Keep ±5% with the final value.

3 Answer in full sentences

The resistor’s resistance is 5,600 ohms. The gold band indicates a tolerance of ±5%.

Every problem needs all three rows. Number each answer, show your calculations, and answer every part in full sentences with units. Do not rewrite the question.

Continue to page 2 to calculate the tolerance range.

Keep this format in your notebook.

What does ±5% mean?

SAME EXAMPLE • 5,600 Ω ±5% • Tolerance range

± means “plus or minus.”

The measured resistance may be above or below the nominal (marked) value by the tolerance amount. Calculate the amount in ohms first, then subtract it and add it.

1 Convert the percentage

5% = 5 ÷ 100 = 0.05

PERCENT = OUT OF 100

Divide the percentage by 100 before multiplying.

2 Find the tolerance amount

5,600 Ω × 0.05 = 280 Ω

THIS IS THE AMOUNT

±5% of 5,600 Ω is ±280 Ω. It is not ±5 Ω.

3 Use minus and plus separately

Lower limit: 5,600 − 280 = 5,320 Ω

Upper limit: 5,600 + 280 = 5,880 Ω

MINUS → LOWER

Subtract 280 Ω.

PLUS → UPPER

Add 280 Ω.

Write the result in full sentences:

The resistor’s resistance is 5,600 ohms with a tolerance of ±5%. Its acceptable resistance range is 5,320 to 5,880 ohms.

4 Check a measured value

Suppose a multimeter reads 5,740 Ω. Compare it with both limits:
5,320 Ω ≤ 5,740 Ω ≤ 5,880 Ω

Full-sentence conclusion: The measured resistance of 5,740 ohms is within tolerance because it lies between 5,320 and 5,880 ohms.

Tolerance work is required: show the percent conversion, multiplication, subtraction, and addition for every problem that asks for a tolerance range.

Keep this format in your notebook.